

# PROTECTIVE DNS

## Layer 4, the Moat: the free filter for your whole home

DNS is the internet's phone book. Point your home network at a safe DNS and it quietly blocks porn, malware, and scam sites for every device on your WiFi, for free. It is the easiest layer to add, and the easiest to forget.

### USE THIS ONE: CIRA CANADIAN SHIELD (FAMILY)

IPV4 PRIMARY

**149.112.121.30**

IPV4 SECONDARY

**149.112.122.30**

IPV6 PRIMARY

**2620:10A:80BB::30**

IPV6 SECONDARY

**2620:10A:80BC::30**

Canadian, run by CIRA (the .ca registry), free. Enter the IPv4 pair; add the IPv6 pair too if your router uses IPv6 (or turn IPv6 off). The Family tier blocks adult content, malware, and phishing. Other options are on the last page.

#### WHAT IT BLOCKS

Pornography, malware, and phishing or scam sites, on every device connected to your home WiFi at once.

#### WHAT IT DOES NOT BLOCK

Social media, YouTube, gaming, or self-harm content. And nothing on cellular data. It is a backstop, not the whole castle.

#### TWO WAYS TO USE THESE NUMBERS

**1. Your router** covers every device on your home WiFi at once (steps below). **2. The free CIRA Canadian Shield app** puts the same protection on one phone or tablet, even on cellular, with no router change at all (page 3). Either works. Many families do both.

## START HERE: FIND YOUR ROUTER

DNS is entered in your router's settings page, which looks different on every box. Answer these first:

- 1. Which ISP?** Telus and Rogers are covered here. Others are similar.
- 2. Is the box black or white?** It tells you the model (see the next pages).
- 3. Open the admin page** in a browser: Telus **192.168.1.254**, Rogers **10.0.0.1**. The login is usually on a sticker on the router.



## WHICH TELUS BOX DO YOU HAVE?

| ROUTER MODEL        | LOOK FOR         | ADMIN PAGE    | CAN YOU CHANGE DNS?     |
|---------------------|------------------|---------------|-------------------------|
| Actiontec T3200M    | Black box        | 192.168.1.254 | Yes, full support       |
| NH20A (Arcadyan)    | White, PureFibre | 192.168.1.254 | Yes, via DHCP           |
| WiFi Hub            | White tower      | 192.168.1.254 | Partial, often bypassed |
| NH20T (Technicolor) | White, PureFibre | 192.168.1.254 | No, greyed out          |
| FXA5000             | White            | 192.168.1.254 | Primary DNS only        |

### ACTIONTEC T3200M (BLACK BOX) - FULL SUPPORT

1. Open a browser and go to **192.168.1.254**. Log in (username admin, password on the sticker).
2. Click **Advanced Setup**, then **I am an Advanced User**.
3. Click **LAN IP Settings**, find **Set the DNS values**.
4. Change DNS Server 1 and 2 from "DNS relay performed by Gateway" to **Statically Assigned**.
5. Enter the two addresses from page 1, click **Apply**.
6. Recommended: in IPv6 LAN Settings, disable ULA Support.

### NH20A (WHITE, ARCADYAN) - WORKS VIA DHCP

1. Open the front panel to find the login, go to **192.168.1.254** and log in.
2. Open the DHCP settings, enter your DNS addresses, and save.

### WIFI HUB (WHITE TOWER) - PROBLEMATIC

This box has a built-in resolver that often ignores custom DNS and pushes itself as the DNS. Try: Network > WAN > Edit, set "Connect to DNS automatically" to No, enter the addresses, Save; disable IPv6. If it does not stick, use the device-level method (page 3) or Bridge Mode with your own router.

### NH20T (TECHNICOLOR) - NOT CHANGEABLE & FXA5000 - PRIMARY ONLY

NH20T DNS fields are greyed out, there is no fix at the router. FXA5000 allows a primary DNS only (Local Network > Show Advanced > DHCP > DNS Server). For NH20T, use the device-level method on page 3, or Bridge Mode with your own router.

## ROGERS GATEWAYS

| MODEL       | LOOK FOR | ADMIN PAGE | CAN YOU CHANGE DNS? |
|-------------|----------|------------|---------------------|
| XB6 (Gen 1) | Black    | 10.0.0.1   | No, locked          |
| XB7 (Gen 2) | White    | 10.0.0.1   | No, locked          |
| XB8 (Gen 3) | White    | 10.0.0.1   | No, locked          |

### ROGERS LOCKS DNS ON EVERY GATEWAY

Rogers has confirmed custom DNS is not available on any Xfinity gateway, through the admin page or the app, with no plans to enable it. The only way is Bridge Mode plus your own router.

### BRIDGE MODE + YOUR OWN ROUTER

1. Go to **10.0.0.1** and log in (admin / password, or the sticker).
2. Go to **Gateway > At a Glance**, find **Bridge Mode**, and enable it.
3. Connect your own router (a TP-Link Deco is ideal) to an ethernet port.
4. Set the DNS on your own router, or in the Deco app.

**Ignite TV caution:** Bridge mode can affect Ignite TV boxes; IPv6 must stay on for them.

## NO ROUTER CHANGE? USE THE FREE APP

If your box is locked (Telus WiFi Hub or NH20T, any Rogers gateway), or you want a phone protected on cellular too, skip the router. Install CIRA's free **Canadian Shield** app on each device, the same protection, on one device at a time.

### IPHONE / IPAD

1. Install **CIRA Canadian Shield** from the App Store (free).
2. Open it and allow the connection profile when asked.
3. Pick a level: **Family** on a child's device blocks adult content too.
4. In Screen Time, restrict VPN changes so it stays on.

### ANDROID

1. Install **CIRA Canadian Shield** from Google Play (free).
2. Open it, turn protection on, allow the connection.
3. Pick **Family** on a child's device.

### WANT MORE CONTROL?

NextDNS is a free, more advanced alternative (App Store, Google Play, or nextdns.io) that lets you build custom block lists per device. Same idea, more dials.

## SAFE DNS OPTIONS

Any of these go in the same DNS fields. CIRA is Canadian-run; CleanBrowsing and OpenDNS filter more aggressively.

| PROVIDER                      | PRIMARY         | SECONDARY       | BEST FOR                         |
|-------------------------------|-----------------|-----------------|----------------------------------|
| CIRA Canadian Shield (Family) | 149.112.121.30  | 149.112.122.30  | Canadian, recommended            |
| CleanBrowsing (Family)        | 185.228.168.168 | 185.228.169.168 | Stronger content filter          |
| OpenDNS FamilyShield          | 208.67.222.123  | 208.67.220.123  | No account needed                |
| Cloudflare for Families       | 1.1.1.3         | 1.0.0.3         | Fastest, blocks adult + malware  |
| Quad9                         | 9.9.9.9         | 149.112.112.112 | Security only, no content filter |

### MAKE THE CHANGE TAKE EFFECT

The easy way: **restart your router, then restart your phones and computers**, that clears the old settings. To clear it directly instead: on a **Mac**, open the Terminal app, type **sudo dscacheutil -flushcache** and press Enter (it may ask for your password). On a **Windows PC**, open Command Prompt, type **ipconfig /flushdns** and press Enter.

### CHECK IT WORKED

On any device on your home WiFi, visit **HomeNetworkCheckup.com**. Your score should jump from around 30% to 70 or 80%. That confirms the filter is live. It does not measure social media or gaming, those need the router and device layers.

### THE HONEST LIMITS

DNS only works on your home WiFi. The moment a phone switches to cellular data, it stops. And it does not touch social media, YouTube, or gaming. Treat it as a free safety net underneath the device controls (Apple Screen Time or Google Family Link) and the router (TP-Link Deco), not a replacement for them.

### KEEP GOING

This is one of five layers. With the device, router, and DNS layers on, put the Safe Family Internet decal in your window so other families know they are not doing this alone.